

Transformer based Time Series Latent Diffusion Model for Bio-signal generation

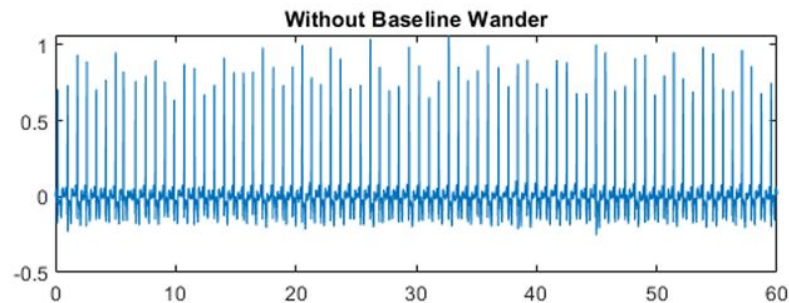
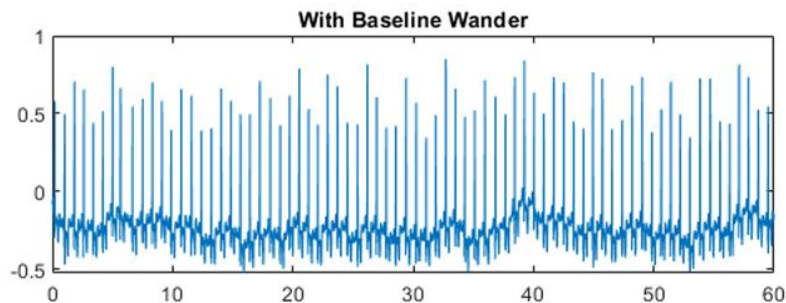
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Problem Setting

- **Motive:** Generate realistic synthetic ECG signals for medical research and model training.
- **Why it matters:**
 - Data privacy: ECG's are biomarkers thus security is important
 - Data scarcity: Low recorded data of anomalous signals
 - Robust model training in healthcare by augmenting unbalanced data
- **Goal:** Models trained on individual classes of ECG data with less samples to generate realistic synthetic signals

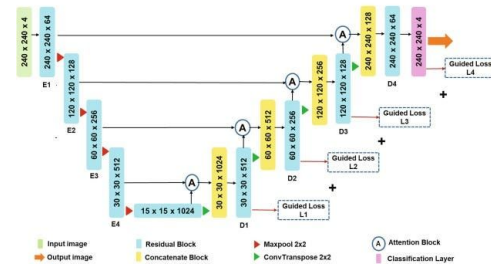
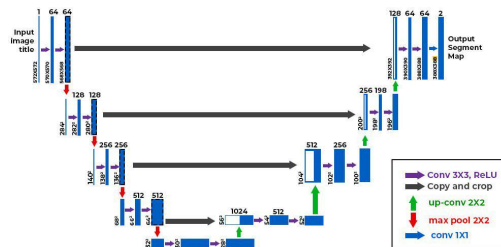
Data and Preprocessing

- Dataset: MIT-BIH arrhythmia dataset from Physiobank; chunked into PQRST beat-bundles
- Raw voltage values have noise known as baseline wander; process and save as python-readable CSV file from MATLAB

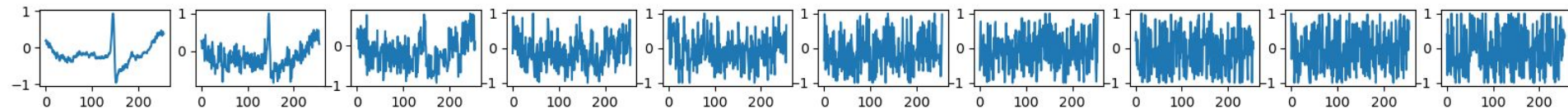
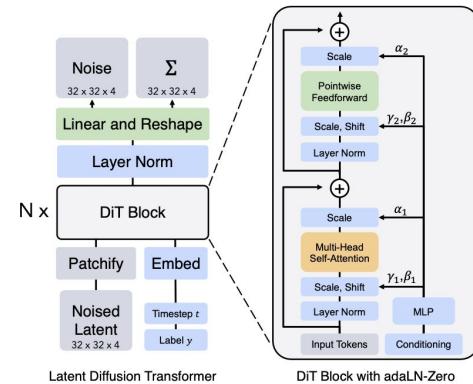


Baseline Model and Innovations

- Baseline Model:
 - Convolution UNet based Latent Diffusion Models.
- Innovations:
 - UNet + Attention based LDM
 - 1D and 2D Diffusion Transformer based LDM



Architecture Image credits: [1] Olaf Ronneberger, Philipp Fischer, & Thomas Brox. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. [2] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Matthias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y Hammerla, Bernhard Kainz, Ben Glocker, & Daniel Rueckert. (2018). Attention U-Net: Learning Where to Look for the Pancreas. [3] William Peebles, & Saining Xie. (2023). Scalable Diffusion Models with Transformers.

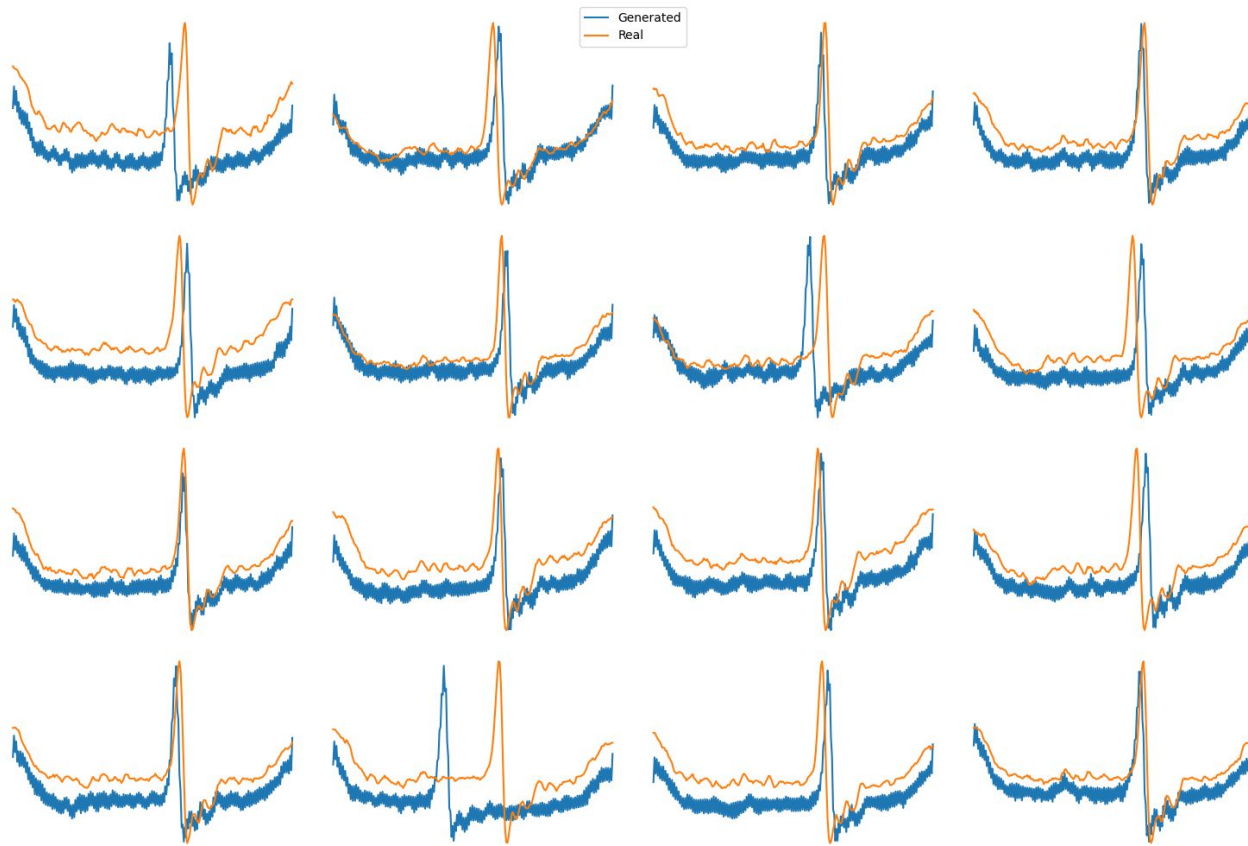


Training Specifics

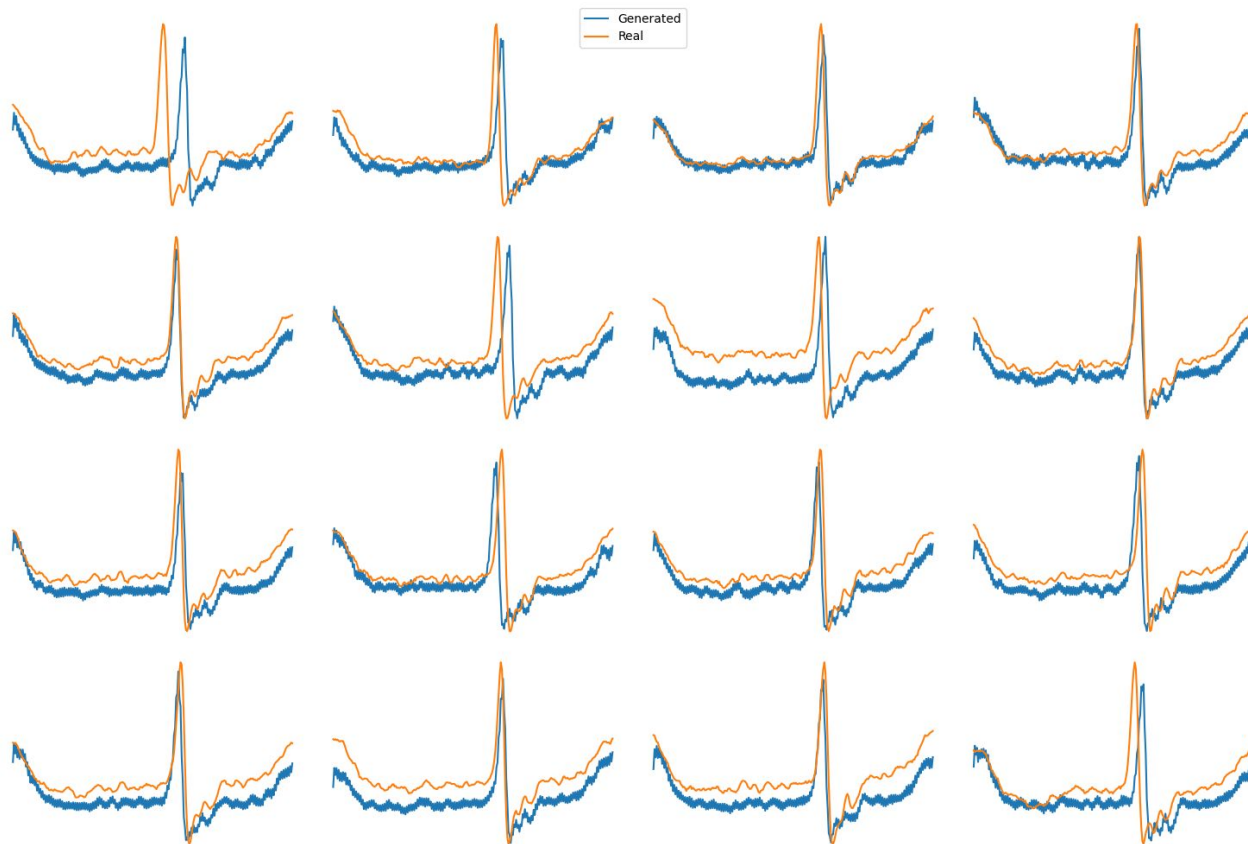
- Each 1D model (UNet, UNet+Attn and DiT) trained for 500 epochs
- Parameters and Training time are as follows:

	Number of Params	Training Time
UNet	18.6M	2:15 minutes
UNet + Attn	20M	2:30 minutes
DiT	6.6M	9:18 minutes

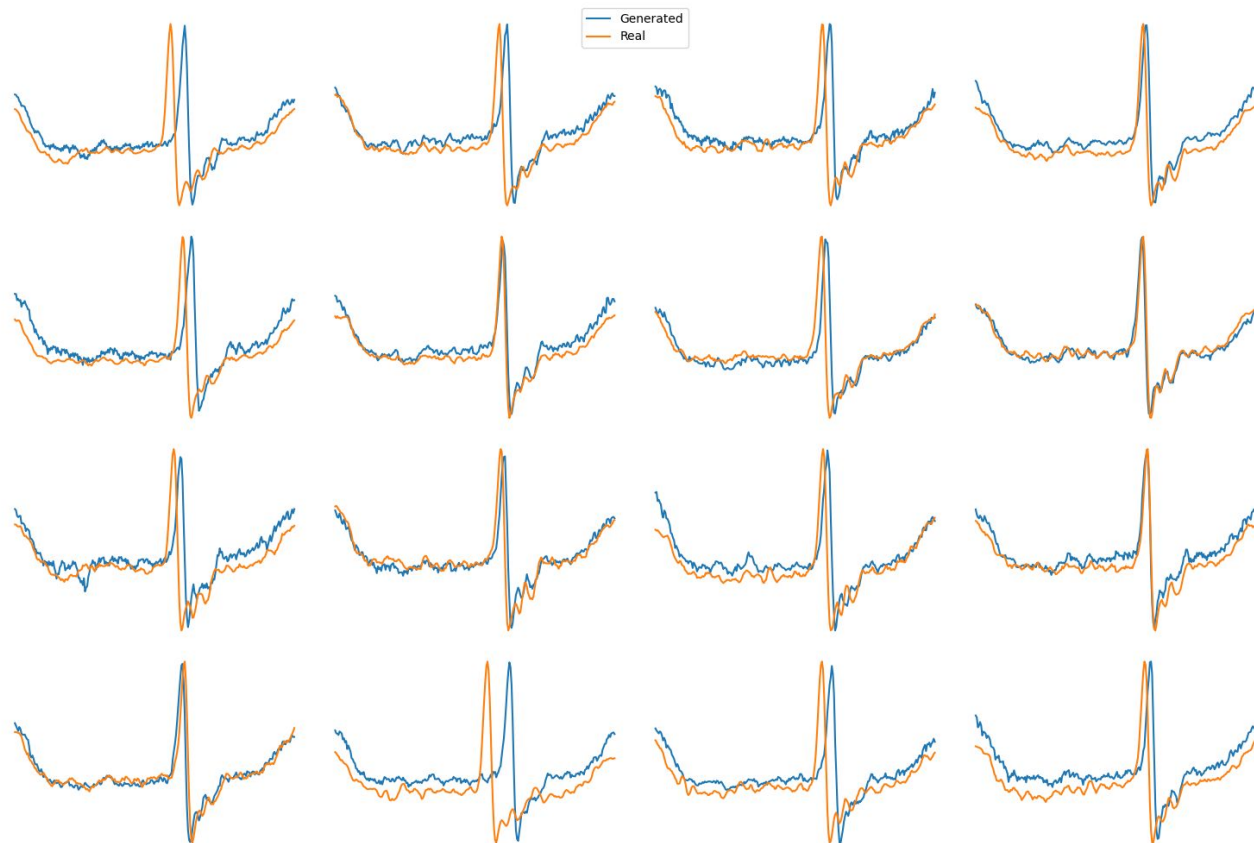
Qualitative Evaluation: Baseline Model



Qualitative Evaluation: UNet + Attention



Qualitative Evaluation: Diffusion Transformers



Quantitative Evaluation

	UNet (Baseline)	UNet + Attention	DiT	Real
↑ UQI	0.976983	0.98397	0.991198	0.992074
↑ VIFP	0.020747	0.014356	0.027133	0.041272
↓ SAM	0.140441	0.123602	0.098887	0.097069
↓ ERGAS	3.509854	3.090588	2.476627	2.432174
↓ RASE	310.643261	261.918316	198.162326	195.699083
↑ SIFT	11.46875	16.40625	21.734375	28.171875
↑ SSIM	0.910642	0.915315	0.930084	0.929843
↓ DTW	33.832653	32.251479	10.942024	9.948521

- Similarity metrics for “S” class of ECG against Real samples
- Overall improvement in metrics compared to Baseline
- [Paper](#)[1] shows similarity metrics can be used to classify with high confidence
- Arrows indicate if higher is better or lower

[1] Singh, A.V., Kumar, I.A., Hota, M.K. (2022). ECG Based Biometric Recognition Using Similarity Measure and Feature Matching. In: Mekhilef, S., Shaw, R.N., Siano, P. (eds) Innovations in Electrical and Electronic Engineering. ICEEE 2022. Lecture Notes in Electrical Engineering, vol 894. Springer, Singapore.
https://doi.org/10.1007/978-981-19-1677-9_52

Quantitative Evaluation

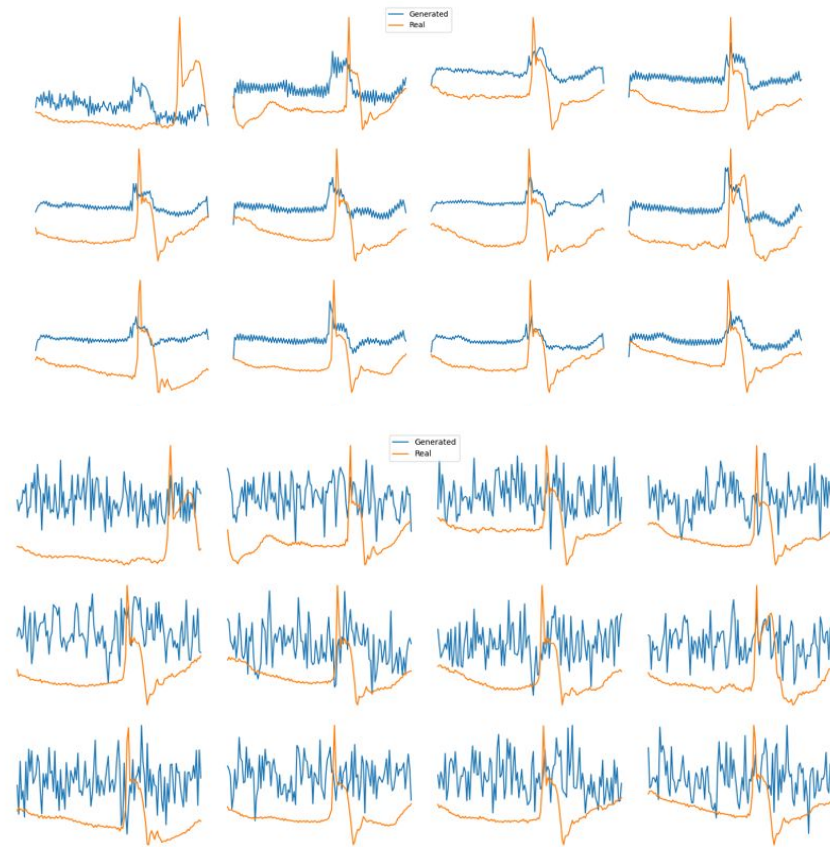
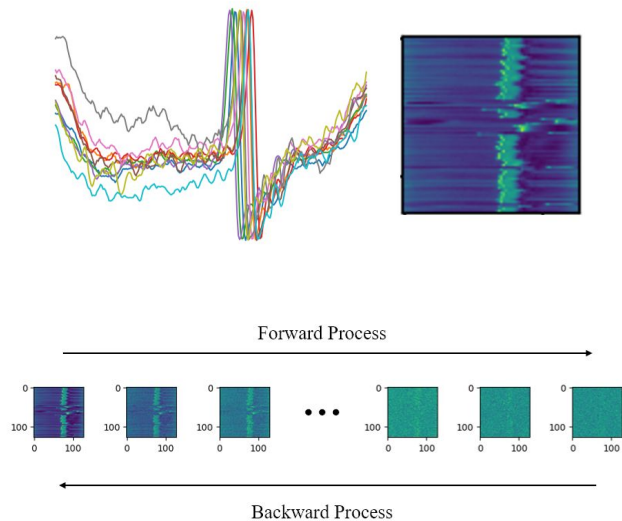
	UNet (Baseline)	UNet + Attention	DiT	Real
↑ UQI	0.981467	0.972355	0.991194	0.991618
↑ VIFP	0.021026	0.020376	0.014725	0.032211
↓ SAM	0.127573	0.146466	0.099059	0.097265
↓ ERGAS	3.189263	3.663008	2.48173	2.436973
↓ RASE	269.653488	321.165366	201.849748	195.198826
↑ SIFT	11.484375	4.53125	19.046875	21.15625
↑ SSIM	0.919327	0.91484	0.928326	0.931696
↓ DTW	22.883154	28.608248	45.942745	17.275514

- Similarity metrics for “Q” class of ECG
- Overall improvement in metrics for DiT compared to Baseline
- Unet + Attn did not perform as well as Unet.
- Performance not as expected as ‘Q’ is a fabricated class in the dataset

Experimental: 2D Training

Input: Stacked signals forming an image

Output: Extracted rows as individual signals



Thank you